Prévention Santé Sécurité

Standard Ergonomics

Reference

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The Ergonomics standards presented in this document aim to provide a practical guide to define the means to deploy to effectively assist the personnel on their daily activities on the project site.

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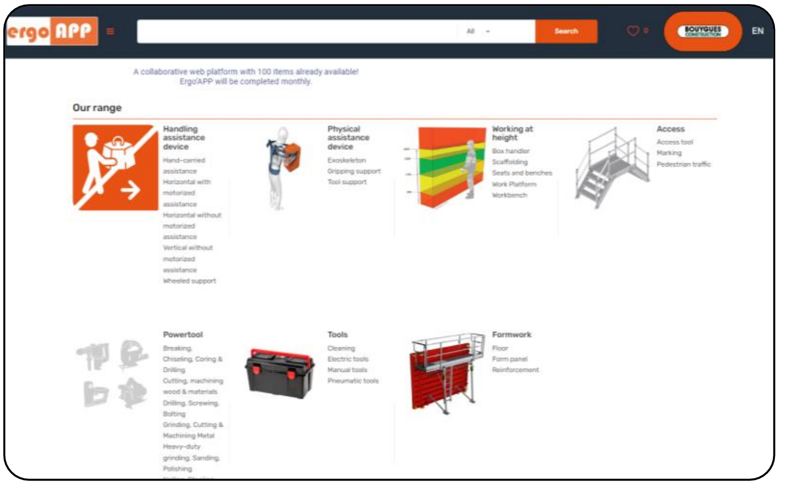
1. The 10 Ergonomics Golden Rules

What is Ergonomics? Ergonomics is a science-based discipline, which fits the task to the worker.

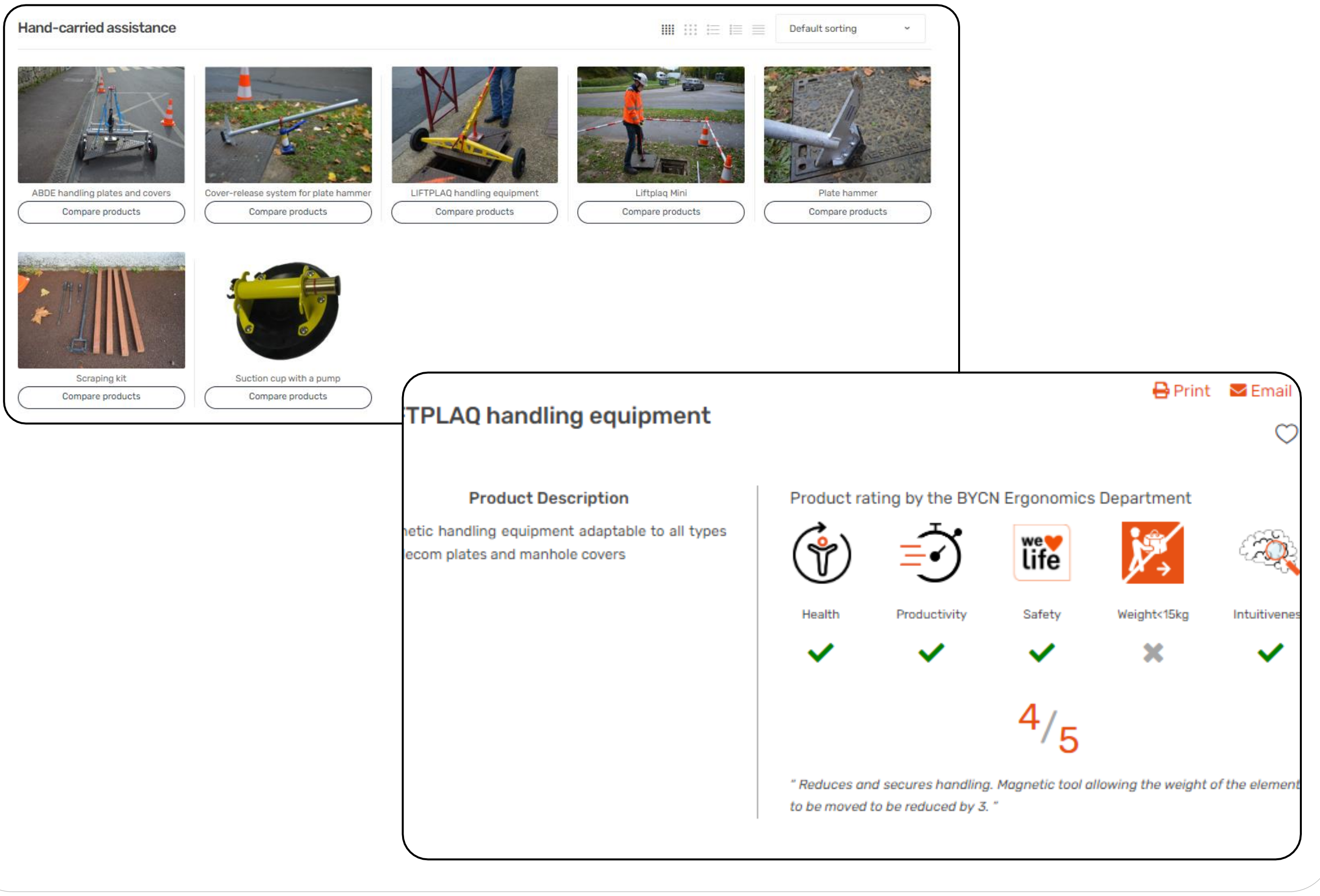
In addition to those golden rules, it is recommended to performed micro breaks between repetitive and strenuous tasks.
These micro breaks can be anywhere between 30 to 60 seconds.

2. Ergo'APP

A collaborative web platform to assist in the selection of ergonomics tools and items is available via the SharePoints of BYCN and CWD.



It contains the list of equipment checked, tested and approved by the Ergonomics teams as well as an evaluation of their added values.



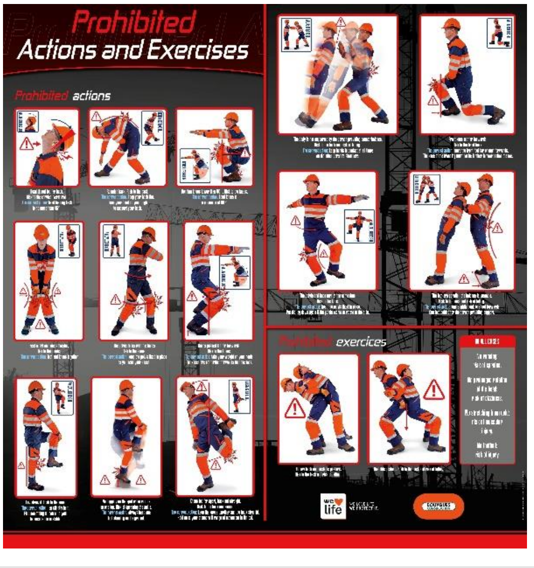
3. Warm-up exercise

This helps in:

- Minimising the risks of physical injury
- Providing a physical and mental “waking-up” (concentration)
- Preparing and improving performance
- Strengthening team cohesion

The warming up exercise is led by:

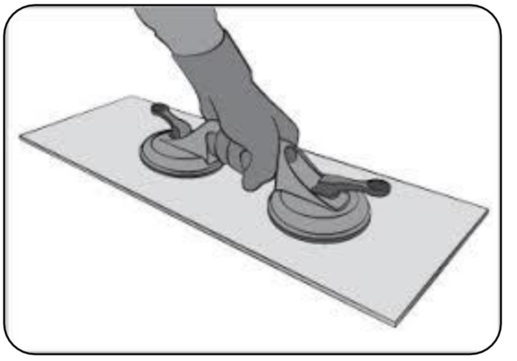
- **A trained employee from the site**
- **Or from an outside agency**



4. Manual lifting accessories

Glass vacuum lifter

Can be used to lift, carry and position sheet materials (glass, metal, etc.) with its secure grip and reduces the risk of accidents (cuts, fall, etc.)



Double pad vacuum lifter

Liftplaq

With its powerful magnet fitting, it can be used to lift and move metal plates

Pipe grabs

Provides safe handling and reduces the risk of falls

With all these tools, manual handling training is still necessary.
Remember to bend with knees.

5. Trolleys

Following-robot

The following-robot accompanies the operators independently and eliminates handling.
Its four-wheel drive and compact dimensions can easily carry up to 300 kg of equipment on all types of terrain.

Drum hand truck

Drum hand truck with safety lock on top

Vacuum trolley

Vacuum lifter for lifting and transporting panels

Stair climbing robot

Stair climbing robot with panel holder
A device that provides safe handling on stairs, eliminating load carrying and reducing the risk of falls. Can be automated or remote controlled without constraints for operators.

Be careful to balance and secure the load to avoid tipping

6. Physical assistive devices

Physical assistive devices are part of the new technologies that offer help at work. They can be used notably for improving productivity and reducing work hardship. Integrated into a global prevention process it can contribute to the prevention of musculoskeletal disorders. These devices should be considered as tools that can be used for specific activities and tasks and not as a PPE.

The exo vest

The exo vest is an exoskeleton-type physical assistive device designed to alleviate strain endured by the operator when working with arms held in the air and over time without forcing or disturbing the movement..

The exo vest supports the arms and can be adjusted accordingly to compensate the weight of the tools up to about 7 kg.

It can be used especially for finishing works involving ceiling sanding or scabbling.

Caution: This is an assistive device to maintain the arms in the air or to compensate for the weight of a tool. It should not be used as a device to assist in carrying a load

Zero G arm

The zero gravity arm is a device that can be used to support a tool as it is being used while simultaneously absorbing some of the vibrations.

The operator no longer has to hold or carry the tool but simply guides it.

Hacking time on reinforced concrete / 2.5 with Zero G arm

7. Bench grinders

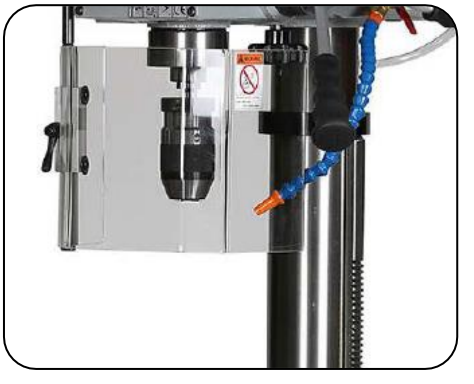
Use of drill presses poses a multitude of hazards, due in particular to the very high-speed rotation of the tool. The machine must be designed and used in accordance with the European Directive 2006/42/EC on machinery or local directive and EN 12717+A1:2009 standard, if there are no stricter local regulations.

Safety requirements for the machine

- The machine **must comply** with the European Directive on machinery and bear the CE marking, if there are no stricter local regulations.
- **Mandatory emergency stop system** on the machine is clearly visible and within easy reach (of "mushroom head" type).
- **Mandatory protective shield or guard** around all moving parts are in position when the machine is in use
- **Lockout /tagout system** is in place to cut off the power supply when carrying out servicing operations on the machine

Setup in the workshop

Any modification of the machine is prohibited



- **Must be bolted to the floor** with 4 bolts for stability
- Metal parts are grounded
- Safe work practices sheet and emergency procedure are posted on the side of the machine
- Provide a non-slip surface in front of the machine to protect against the risk of slipping on metal cuttings and lubricant splashes

Protective shield must be in place when machine is in use

For the user

- Train the operator on how to use the machine with technical instructions
- Certification is issued by the trainer and authorization by the Project Manager
- Do not wear loose or wide-sleeved clothing, jewelry, bracelets, watches… Wear cap or hair net for long hair
- Clean the machine regularly **with vacuum and brush** when it is not running and switched off
- Wear appropriate PPE, **no gloves when machining and wear tight-fitting clothing**


Safety eyewear


Hearing protection


Appropriate and tight-fitting work clothing


Gloves when not machining

8. Bench grinders

Use of bench grinder poses a multitude of hazards, due in particular to the high-speed rotation of the grinding wheels. The machine must be designed and used in accordance with the European Directive 2006/42/EC on machinery or local directive and EN 61029-2-4 standard, if there are no stricter local regulations.

Safety requirements for the machine

- The machine **must comply** with the European Directive on machinery and bear the CE marking , if there are no stricter local regulations.
- **Mandatory emergency stop system with electronic brake** on the machine is clearly visible and within easy reach (of "mushroom head" type)
- **Mandatory spark arrester or tongue guard and transparent protective shield** are in position
- **Lockout /tagout system** is in place to cut off the power supply when carrying out servicing operations on the machine
- Give preference to machines with **integrated dust extraction**

Any modification of the machine is prohibited

Setup in the workshop

- **Must be mounted on a pedestal or bench, if mounted on a pedestal, the pedestal must also be bolted to the floor** with 4 bolts
- **Adjust** tool rests to **within 3 mm** of wheels
- **Maintain a maximum of 5 mm wheel exposure** with a tongue guard
- Metal parts are grounded
- Safe work practices sheet and emergency procedure are posted on the side of the machine

Protective shield must be in place when machine is used

For the user

- Train the operator on how to use the machine with technical instructions
- For least exertion/strain, user must control machine at their elbow height; such that the elbows are aligned with the working base.
- Certification is issued by the trainer and authorisation by the Project Manager
- Wear appropriate PPE, **no gloves when machining and wear tight-fitting clothing**


Safety eyewear


Hearing protection


Appropriate and tight-fitting work clothing


Gloves when not machining

9. Drilling robot

Preferably use of drilling robots under the following configurations:

- High number of drills
- Large bore diameter
- Significant bore depth

10. Guide rail for drilling

Advantages:

- Less strenuous
- Higher productivity
- Better safety

	Guide rail	Manual drilling
Posture	Work standing, body upright No neck strain Minor lower limb stresses Operator to take short micro breaks and consider job rotation	Bending the torso Bending the lower limbs
Exertion	Low exertion (transmission by the remote control system)	High exertion and requiring the need to stay in a static position
Supporting the load	The equipment is held by the guide rail, so no load to support	Holding the electric power tool in a static position
Drilling speed	Able to maintain the same pace throughout the day	Fatigue leading to reduced pace